GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

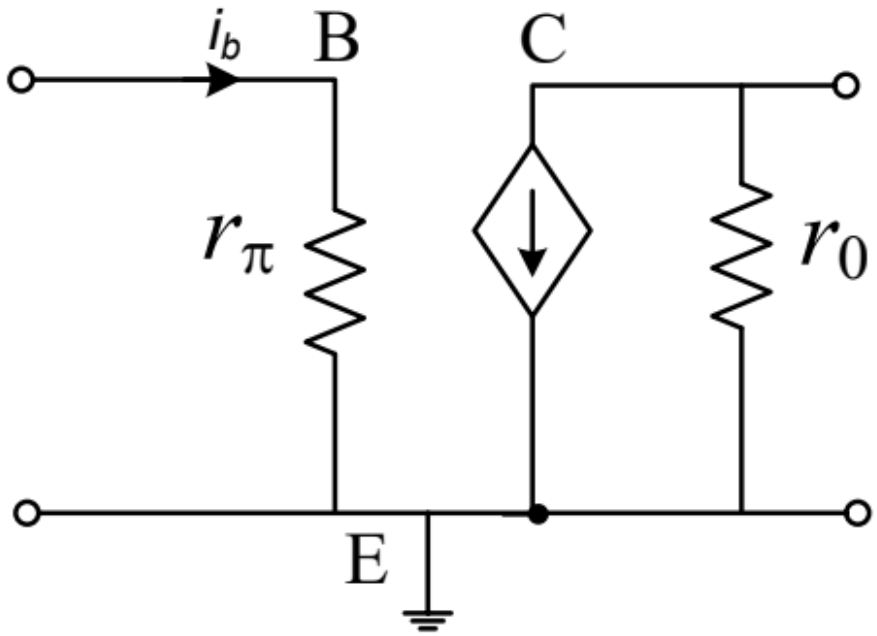
Typeset edition

Papers included: GATE 2012.

GATE 2012

GATE 2012 | Q.1 | 1 mark

The current i_b through the base of a silicon npn transistor is $1 + 0.1\cos(10000\pi t)$ mA. At 300 K, the r_π in the small signal model of the transistor is



(A) 250 Ω
(B) 27.5 Ω
(C) 25 Ω
(D) 22.5 Ω

Answer: (C) [answer matches official key]

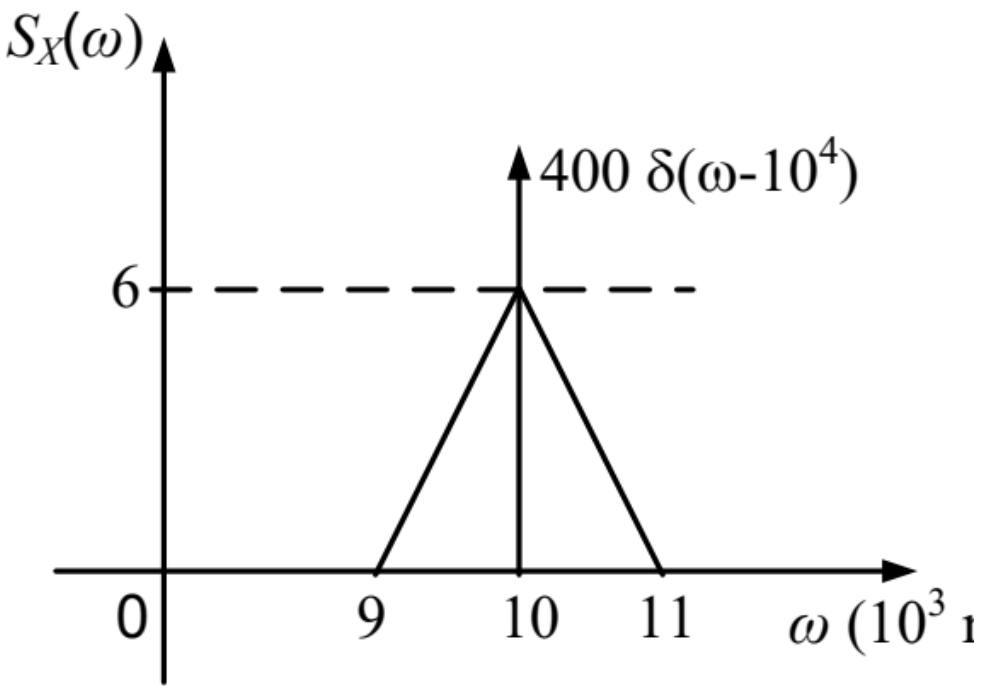
Solution. Step 1. The dc component of the base current is $I_B = 1$ mA, which sets the bias.

Step 2. $r_\pi = \dfrac{V_T}{I_B} = \dfrac{\beta V_T}{I_C} = \dfrac{25\text{ mV}}{1\text{ mA}} = 25\ \Omega$ at 300 K ($V_T \approx 25$ mV).

GATE 2012 | Q.2 | 1 mark

The power spectral density of a real process $X(t)$ for positive frequencies is shown below. The values of $E[X^2(t)]$ and $|E[X(t)]|$, respectively, are

ity of a real process $X(t)$ for positive $|[X(t)]|$, respectively, are



(A) $6000/\pi,\ 0$
(B) $6400/\pi,\ 0$
(C) $6400/\pi,\ 20/(\pi\sqrt{2})$
(D) $6000/\pi,\ 20/(\pi\sqrt{2})$

Answer: (B) [answer matches official key]

Solution. Step 1. For a real process the PSD is even, so the total power is $\frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(\omega)\,d\omega = \frac{1}{2\pi}\cdot 2\int_0^{\infty} S_X\,d\omega$.
Step 2. Positive-frequency part: triangle of base 2×10^3 and height 6 (area 6000) plus the impulse $400\,\delta(\omega-10^4)$ (area 400): total 6400.
Step 3. $E[X^2] = \frac{1}{2\pi}\cdot 2\cdot 6400 = \frac{6400}{\pi}$.
Step 4. There is no impulse at $\omega = 0$, so the dc value (mean) is 0.

GATE 2012 | Q.3 | 1 mark

In a baseband communications link, frequencies up to 3500 Hz are used for signaling. Using a raised cosine pulse with 75% excess bandwidth and for no inter-symbol interference, the maximum possible signaling rate in symbols per second is

(A) 1750
(B) 2625
(C) 4000
(D) 5250

Answer: (C) [answer matches official key]

Solution. Step 1. Raised-cosine bandwidth: $B = \frac{1+\alpha}{2T}$, where $1/T$ is the symbol rate.
Step 2. $3500 = \frac{1.75}{2T} \Rightarrow \frac{1}{T} = \frac{2\times 3500}{1.75} = 4000$ symbols/s.

GATE 2012 | Q.4 | 1 mark

A plane wave propagating in air with $\vec{E} = (8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z)e^{j(\omega t+3x+4y)}$ V/m is incident on a perfectly conducting slab positioned at $x \leq 0$. The $\vec{E}$ field of the reflected wave is

(A) $(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t+3x+4y)}$ V/m
(B) $(-8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z)e^{j(\omega t+3x+4y)}$ V/m
(C) $(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t-3x-4y)}$ V/m
(D) $(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t-3x+4y)}$ V/m

Answer: Marks to all [answer matches official key]

Solution. Step 1. Tangential components at $x = 0$ (y, z) must cancel, so the reflected y, z amplitudes are $-6, -5$.
Step 2. The x-component of the wave vector reverses $(+3x \to -3x)$ while the y variation e^{j4y} is unchanged (phase matching).
Step 3. The reflected wave must be transverse, $\vec{E}_r \cdot \vec{k}_r = 0$ with $\vec{k}_r \propto (3, -4, 0)$: $E_{rx} \cdot 3 + E_{ry} \cdot (-4) = 0 \Rightarrow E_{rx} = -8$.
Step 4. $\vec{E}_r = (-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t-3x+4y)}$, option (D). (The official key awarded marks to all for this question.)

GATE 2012 | Q.5 | 1 mark

The electric field of a uniform plane electromagnetic wave in free space, along the positive x direction, is given by $\vec{E} = 10(\hat{a}_y + j\hat{a}_z)e^{-j25x}$. The frequency and polarization of the wave, respectively, are

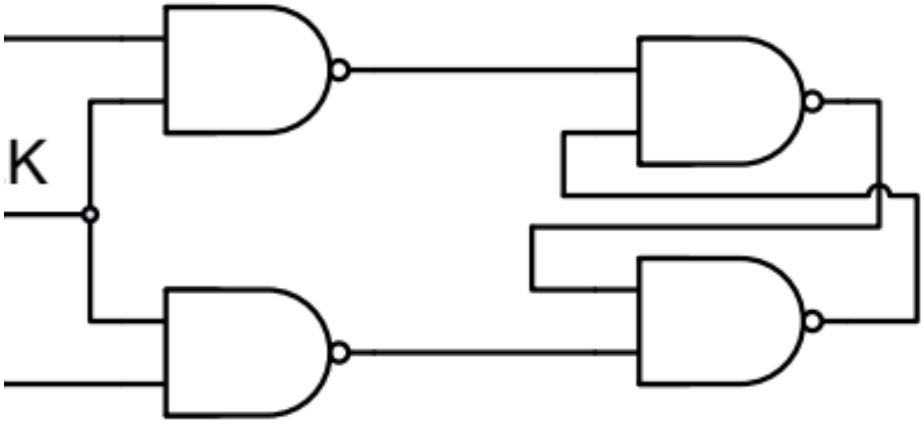
(A) 1.2 GHz and left circular
(B) 4 Hz and left circular
(C) 1.2 GHz and right circular
(D) 4 Hz and right circular

Answer: (A) [answer matches official key]

Solution. Step 1. $\beta = 25$ rad/m, so $\omega = \beta c = 25 \times 3 \times 10^8 = 7.5 \times 10^9$ rad/s and $f = \dfrac{\omega}{2\pi} = 1.19 \approx 1.2$ GHz.
Step 2. Equal amplitudes with a 90° phase difference: circular polarization. In the time domain at $x = 0$: $E_y = 10\cos\omega t$, $E_z = -10\sin\omega t$.
Step 3. The vector rotates from $+y$ towards $-z$, which is clockwise about the $+x$ axis of propagation, i.e. left-handed (IEEE convention).

GATE 2012 | Q.6 | 1 mark

Consider the given circuit. In this circuit, the race around



(A) does not occur
(B) occurs when CLK = 0

(C) occurs when CLK = 1 and A = B = 1
(D) occurs when CLK = 1 and A = B = 0

Answer: (A) [answer matches official key]

Solution. Step 1. The circuit is a clocked SR-type latch built from NAND gates; the outputs are fed back only to the output (storage) gates, not to the input steering gates.
Step 2. Race-around occurs when both inputs are 1 and the output toggles repeatedly during the clock pulse (JK latch). Here that toggling path does not exist, so race-around does not occur.

GATE 2012 | Q.7 | 1 mark

The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B. The number of combinations for which the output is logic 1, is

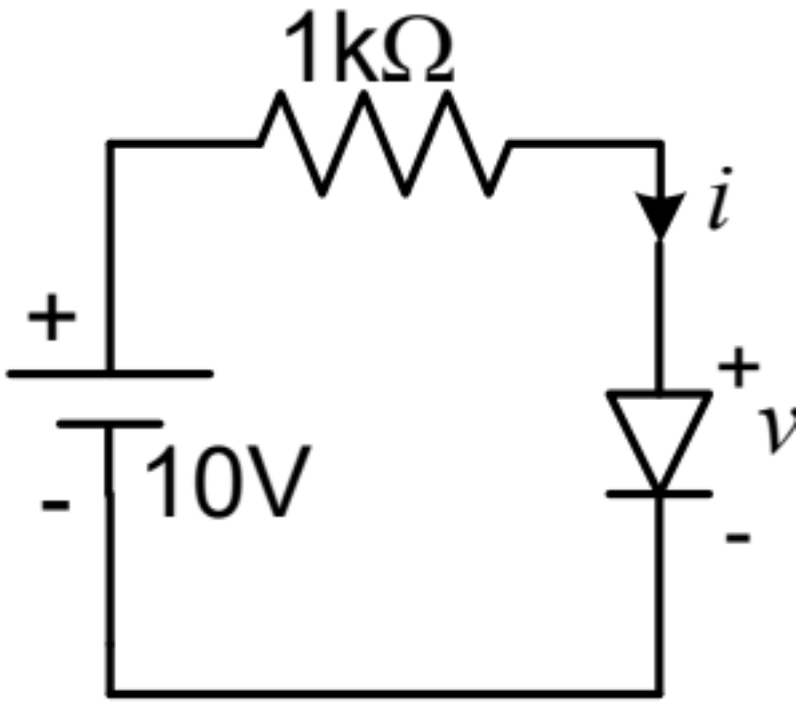
(A) 4
(B) 6
(C) 8
(D) 10

Answer: (B) [answer matches official key]

Solution. Step 1. There are $4 \times 4 = 16$ input combinations.
Step 2. $A > B$, $A = B$ and $A < B$: 4 combinations have $A = B$, and by symmetry $A > B$ and $A < B$ share the remaining 12 equally: 6.

GATE 2012 | Q.8 | 1 mark

The i-v characteristics of the diode in the circuit given below are $i = \begin{cases} \dfrac{v - 0.7}{500} \text{ A}, & v \geq 0.7 \text{ V} \\ 0 \text{ A}, & v < 0.7 \text{ V} \end{cases}$. The current in the circuit is



(A) 10 mA

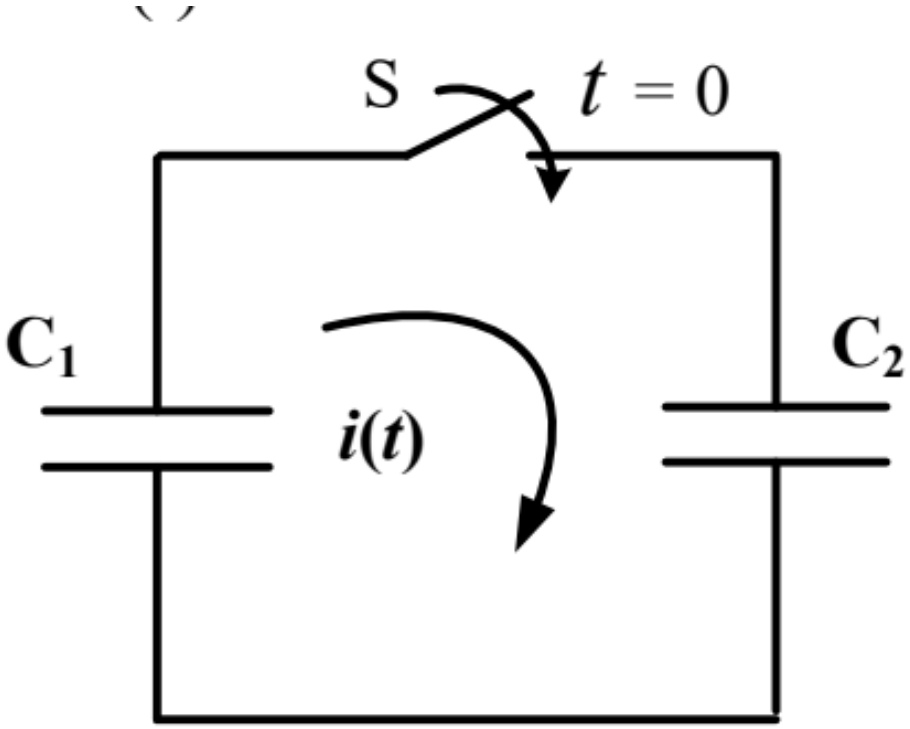
(B) 9.3 mA
(C) 6.67 mA
(D) 6.2 mA

Answer: (D) [answer matches official key]

Solution. Step 1. KVL: $10 = 1000\,i + v$.
Step 2. For $v \geq 0.7$: $v = 500\,i + 0.7$.
Step 3. $10 = 1000i + 500i + 0.7 \Rightarrow i = \dfrac{9.3}{1500} = 6.2$ mA.

GATE 2012 | Q.9 | 1 mark

In the following figure, C_1 and C_2 are ideal capacitors. C_1 has been charged to 12 V before the ideal switch S is closed at $t = 0$. The current $i(t)$ for all t is



(A) zero
(B) a step function
(C) an exponentially decaying function
(D) an impulse function

Answer: (D) [answer matches official key]

Solution. Step 1. The loop contains only ideal capacitors (no resistance) and a switch.
Step 2. Closing the switch forces the capacitor voltages to equalise instantly; a finite charge moves in zero time, which requires an infinite current for an infinitesimal time: an impulse.

GATE 2012 | Q.10 | 1 mark

The average power delivered to an impedance $(4 - j3)\ \Omega$ by a current $5\cos(100\pi t + 100)$ A is
(A) 44.2 W
(B) 50 W

(C) 62.5 W
(D) 125 W

Answer: (B) [answer matches official key]

Solution. Step 1. Only the resistive part dissipates power: $R = 4\ \Omega$.
Step 2. $P = \frac{1}{2} I_m^2 R = \frac{1}{2}(25)(4) = 50$ W.

GATE 2012 | Q.11 | 1 mark

The unilateral Laplace transform of $f(t)$ is $\dfrac{1}{s^2+s+1}$. The unilateral Laplace transform of $t\,f(t)$ is

(A) $-\dfrac{s}{(s^2+s+1)^2}$
(B) $-\dfrac{2s+1}{(s^2+s+1)^2}$
(C) $\dfrac{s}{(s^2+s+1)^2}$
(D) $\dfrac{2s+1}{(s^2+s+1)^2}$

Answer: (D) [answer matches official key]

Solution. Step 1. Property: $\mathcal{L}\{tf(t)\} = -\dfrac{d}{ds}F(s)$.
Step 2. $-\dfrac{d}{ds}\dfrac{1}{s^2+s+1} = \dfrac{2s+1}{(s^2+s+1)^2}$.

GATE 2012 | Q.12 | 1 mark

With initial condition $x(1) = 0.5$, the solution of the differential equation, $t\dfrac{dx}{dt} + x = t$ is

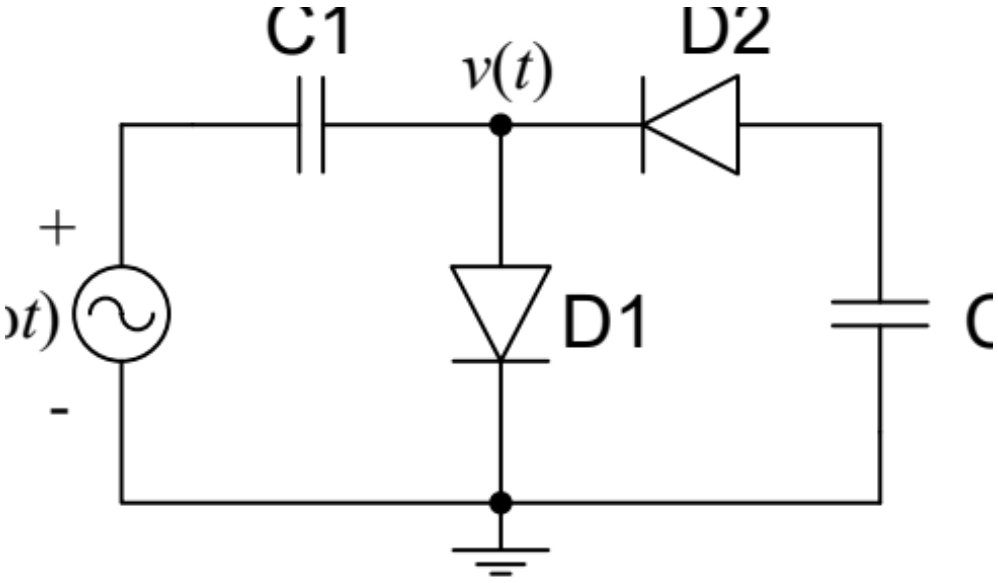
(A) $x = t - \dfrac{1}{2}$
(B) $x = t^2 - \dfrac{1}{2}$
(C) $x = \dfrac{t^2}{2}$
(D) $x = \dfrac{t}{2}$

Answer: (D) [answer matches official key]

Solution. Step 1. The left side is an exact derivative: $\dfrac{d}{dt}(tx) = t$.
Step 2. Integrate: $tx = \dfrac{t^2}{2} + c$.
Step 3. $x(1) = 0.5 \Rightarrow 0.5 = 0.5 + c \Rightarrow c = 0$, so $x = \dfrac{t}{2}$.

GATE 2012 | Q.13 | 1 mark

The diodes and capacitors in the circuit shown are ideal. The voltage $v(t)$ across the diode D_1 is



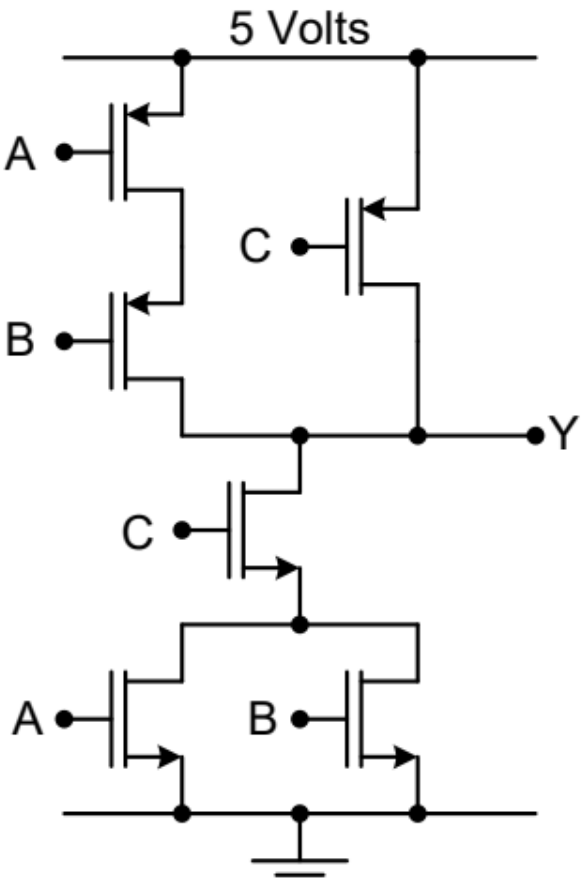
(A) $\cos(\omega t) - 1$
(B) $\sin(\omega t)$
(C) $1 - \cos(\omega t)$
(D) $1 - \sin(\omega t)$

Answer: (A) [answer matches official key]

Solution. Step 1. On the first negative peak of the source (-1 V) diode D_2 conducts and C_1 charges to 1 V (positive on the source side), and it then holds this value.
Step 2. Diode D_1 therefore sees $v(t) = \cos\omega t - 1$, which is never positive, i.e. the waveform is clamped with its peaks at 0 V.

GATE 2012 | Q.14 | 1 mark

In the circuit shown, Y is



(A) $Y = \overline{A}\,\overline{B} + \overline{C}$
(B) $Y = (A + B)C$
(C) $Y = (\overline{A} + \overline{B})\overline{C}$
(D) $Y = AB + C$

Answer: (A) [answer matches official key]

Solution. Step 1. Pull-down network (nMOS): A and B in parallel, in series with C: conducts when $(A + B)C = 1$.
Step 2. A CMOS gate output is the complement of its pull-down function: $Y = \overline{(A + B)C}$.
Step 3. $Y = \overline{A + B} + \overline{C} = \overline{A}\,\overline{B} + \overline{C}$.

GATE 2012 | Q.15 | 1 mark

A source alphabet consists of N symbols with the probability of the first two symbols being the same. A source encoder increases the probability of the first symbol by a small amount ε and decreases that of the second by ε. After encoding, the entropy of the source

(A) increases
(B) remains the same
(C) increases only if $N = 2$
(D) decreases

Answer: (D) [answer matches official key]

Solution. Step 1. The entropy is maximised by the most uniform distribution; the first two probabilities were equal.
Step 2. Making them unequal $(p + \varepsilon,\ p - \varepsilon)$ with the sum fixed reduces $-p \log p$ summed over them (concavity of the entropy), so the entropy decreases.

GATE 2012 | Q.16 | 1 mark

A coaxial cable with an inner diameter of 1 mm and outer diameter of 2.4 mm is filled with a dielectric of relative permittivity 10.89. Given $\mu_0 = 4\pi \times 10^{-7}$ H/m, $\varepsilon_0 = \dfrac{10^{-9}}{36\pi}$ F/m, the characteristic impedance of the cable is

(A) 330 Ω
(B) 100 Ω
(C) 143.3 Ω
(D) 43.4 Ω

Answer: Marks to all [answer matches official key]

Solution. Step 1. $Z_0 = \dfrac{1}{2\pi}\sqrt{\dfrac{\mu_0}{\varepsilon_0\varepsilon_r}}\ln\dfrac{b}{a} = \dfrac{377}{2\pi\sqrt{10.89}}\ln 2.4 = 18.2 \times 0.875 \approx 15.9\ \Omega$.

Step 2. None of the printed options equals this value, so the question was flawed and marks were awarded to all candidates.

GATE 2012 | Q.17 | 1 mark

The radiation pattern of an antenna in spherical co-ordinates is given by $F(\theta) = \cos^4\theta;\ 0 \le \theta \le \pi/2$. The directivity of the antenna is

(A) 10 dB
(B) 12.6 dB
(C) 11.5 dB
(D) 18 dB

Answer: (B) [answer matches official key]

Solution. Step 1. $F(\theta)$ is the field pattern, so the power pattern is $F^2 = \cos^8\theta$.

Step 2. $D = \dfrac{4\pi F_{max}^2}{\int F^2 d\Omega} = \dfrac{4\pi}{2\pi\int_0^{\pi/2}\cos^8\theta\sin\theta\,d\theta} = \dfrac{4\pi}{2\pi/9} = 18$.

Step 3. $10\log_{10}18 = 12.6$ dB.

GATE 2012 | Q.18 | 1 mark

If $x[n] = (1/3)^{|n|} - (1/2)^n u[n]$, then the region of convergence (ROC) of its Z-transform in the Z-plane will be

(A) $\frac{1}{3} < |z| < 3$
(B) $\frac{1}{3} < |z| < \frac{1}{2}$
(C) $\frac{1}{2} < |z| < 3$
(D) $\frac{1}{3} < |z|$

Answer: (C) [answer matches official key]

Solution. Step 1. $(1/3)^{|n|}$ is two-sided: ROC $\dfrac{1}{3} < |z| < 3$.

Step 2. $(1/2)^n u[n]$ is causal: ROC $|z| > \dfrac{1}{2}$.

Step 3. The ROC of the sum is the intersection: $\dfrac{1}{2} < |z| < 3$.

GATE 2012 | Q.19 | 1 mark

In the sum of products function $f(X, Y, Z) = \sum(2, 3, 4, 5)$, the prime implicants are

(A) $\overline{X}Y,\ X\overline{Y}$
(B) $\overline{X}Y,\ X\overline{Y}\,\overline{Z},\ X\overline{Y}Z$
(C) $\overline{X}Y\overline{Z},\ \overline{X}YZ,\ X\overline{Y}$

(D) $\overline{X}Y\overline{Z},\ \overline{X}YZ,\ X\overline{Y}\,\overline{Z},\ X\overline{Y}Z$

Answer: (A) [answer matches official key]

Solution. Step 1. Minterms: $2 = 010$, $3 = 011$, $4 = 100$, $5 = 101$.
Step 2. Pair $(2, 3)$: $\overline{X}Y$. Pair $(4, 5)$: $X\overline{Y}$. No further grouping is possible.
Step 3. The prime implicants are $\overline{X}Y$ and $X\overline{Y}$.

GATE 2012 | Q.20 | 1 mark

A system with transfer function $G(s) = \dfrac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$ is excited by $\sin(\omega t)$. The steady-state output of the system is zero at

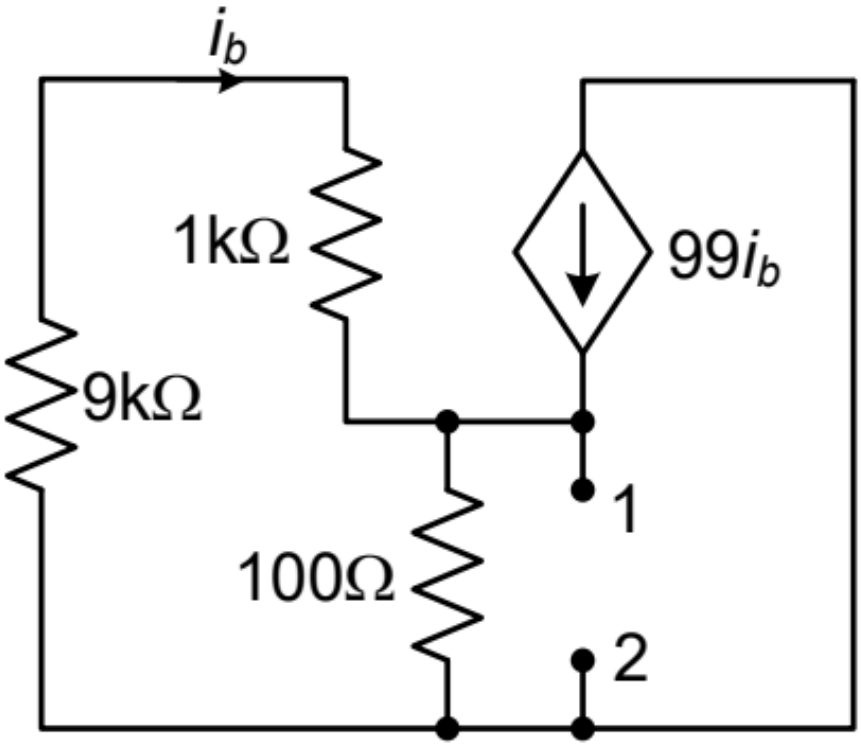
(A) $\omega = 1$ rad/s
(B) $\omega = 2$ rad/s
(C) $\omega = 3$ rad/s
(D) $\omega = 4$ rad/s

Answer: (C) [answer matches official key]

Solution. Step 1. The steady-state output amplitude is $|G(j\omega)|$.
Step 2. $G(j\omega) = 0$ when the numerator vanishes: $s^2 + 9 = 0 \Rightarrow s = \pm j3$.
Step 3. So the output is zero at $\omega = 3$ rad/s.

GATE 2012 | Q.21 | 1 mark

The impedance looking into nodes 1 and 2 in the given circuit is



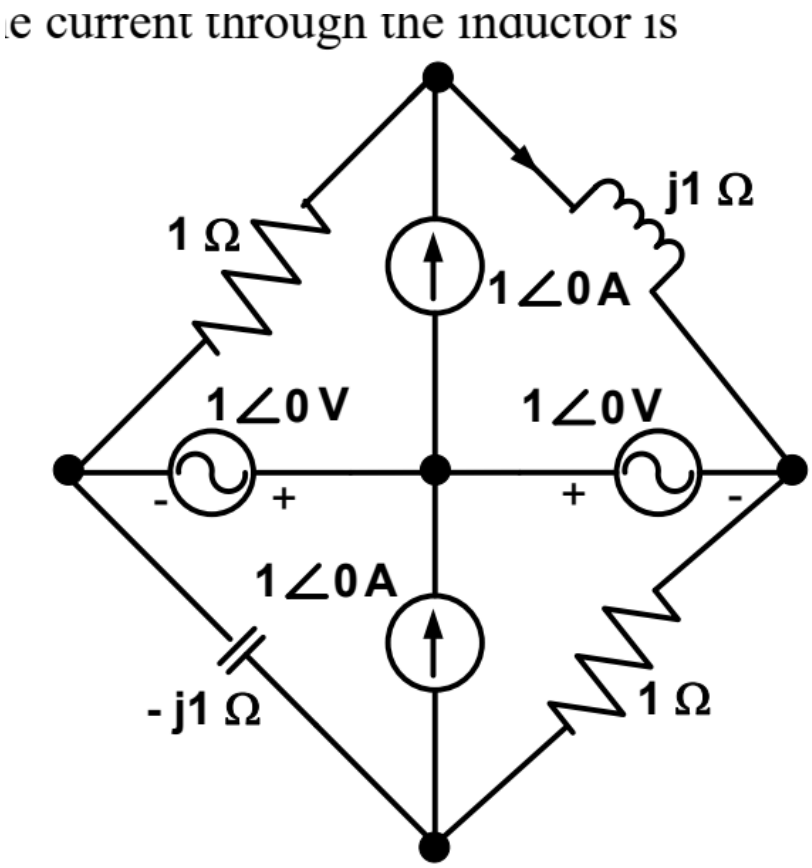
(A) 50 Ω
(B) 100 Ω
(C) 5 kΩ
(D) 10.1 kΩ

Answer: (A) [answer matches official key]

Solution. Step 1. Apply a test current I into node 1 (out of node 2) and let $V = V_1 - V_2$.
Step 2. The 9 kΩ and 1 kΩ resistors form a series path from node 2 to node 1 carrying i_b: $V_2 - V_1 = 10{,}000\, i_b$, i.e. $V = -10{,}000\, i_b$.
Step 3. KCL at node 1: $I + i_b + 99 i_b = \dfrac{V}{100} \Rightarrow I = \dfrac{V}{100} - 100 i_b = \dfrac{V}{100} + \dfrac{V}{100} = \dfrac{V}{50}$.
Step 4. $Z_{in} = \dfrac{V}{I} = 50\ \Omega$.

GATE 2012 | Q.22 | 1 mark

In the circuit shown below, the current through the inductor is



(A) $\dfrac{2}{1+j}$ A
(B) $\dfrac{-1}{1+j}$ A
(C) $\dfrac{1}{1+j}$ A
(D) 0 A

Answer: (C) [answer matches official key]

Solution. Step 1. Take the central node as reference (0 V). The two $1\angle 0^\circ$ V sources fix the left and right nodes at $V_L = V_R = -1$ V.
Step 2. KCL at the top node: the $1\angle 0^\circ$ A source injects 1 A: $1 = \dfrac{V_T - V_L}{1} + \dfrac{V_T - V_R}{j} = (V_T + 1)\left(1 + \dfrac{1}{j}\right) = (V_T + 1)(1 - j)$.
Step 3. Inductor current $= \dfrac{V_T - V_R}{j} = \dfrac{V_T + 1}{j} = \dfrac{1}{j(1-j)} = \dfrac{1}{j+1}$ A.

GATE 2012 | Q.23 | 1 mark

Given $f(z) = \dfrac{1}{z+1} - \dfrac{2}{z+3}$. If C is a counterclockwise path in the z-plane such that $|z+1| = 1$, the value of $\dfrac{1}{2\pi j}\oint_C f(z)\,dz$ is

(A) -2
(B) -1
(C) 1
(D) 2

Answer: (C) [answer matches official key]

Solution. Step 1. The circle $|z+1| = 1$ is centred at -1 with radius 1: it encloses the pole $z = -1$ but not $z = -3$ (distance 2).

Step 2. The residue at $z = -1$ is 1, so $\dfrac{1}{2\pi j}\oint f\,dz = 1$.

GATE 2012 | Q.24 | 1 mark

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

(A) 3/4
(B) 9/16
(C) 1/4
(D) 2/3

Answer: (B) [answer matches official key]

Solution. Step 1. $\max(X, Y) < \frac{1}{2} \iff X < \frac{1}{2}$ and $Y < \frac{1}{2}$.

Step 2. $P(X < \frac{1}{2}) = \dfrac{1/2 - (-1)}{2} = \dfrac{3}{4}$ and likewise for Y.

Step 3. By independence $P = \dfrac{3}{4} \cdot \dfrac{3}{4} = \dfrac{9}{16}$.

GATE 2012 | Q.25 | 1 mark

If $x = \sqrt{-1}$, then the value of x^x is

(A) $e^{-\pi/2}$
(B) $e^{\pi/2}$
(C) x
(D) 1

Answer: (A) [answer matches official key]

Solution. Step 1. $x = i = e^{i\pi/2}$, so $\ln i = \dfrac{i\pi}{2}$.

Step 2. $x^x = e^{x \ln x} = e^{i \cdot i\pi/2} = e^{-\pi/2}$.

GATE 2012 | Q.26 | 2 marks

The source of a silicon ($n_i = 10^{10}$ per cm^3) n-channel MOS transistor has an area of 1 sq μm and a depth of 1 μm. If the dopant density in the source is 10^{19}/cm^3, the number of holes in the source region with the above volume is approximately

(A) 10^7
(B) 100
(C) 10
(D) 0

Answer: (D) [answer matches official key]

Solution. Step 1. Hole density in the heavily doped n-type source: $p = \dfrac{n_i^2}{N_D} = \dfrac{10^{20}}{10^{19}} = 10 \text{ cm}^{-3}$.
Step 2. Volume $= 1\ \mu\text{m}^2 \times 1\ \mu\text{m} = 10^{-12}\ \text{cm}^3$.
Step 3. Number of holes $= 10 \times 10^{-12} = 10^{-11} \approx 0$.

GATE 2012 | Q.27 | 2 marks

A BPSK scheme operating over an AWGN channel with noise power spectral density of $N_0/2$, uses equiprobable signals $s_1(t) = \sqrt{\dfrac{2E}{T}}\sin(\omega_c t)$ and $s_2(t) = -\sqrt{\dfrac{2E}{T}}\sin(\omega_c t)$ over the symbol interval $(0, T)$. If the local oscillator in a coherent receiver is ahead in phase by 45° with respect to the received signal, the probability of error in the resulting system is

(A) $Q\left(\sqrt{\dfrac{2E}{N_0}}\right)$
(B) $Q\left(\sqrt{\dfrac{E}{N_0}}\right)$
(C) $Q\left(\sqrt{\dfrac{E}{2N_0}}\right)$
(D) $Q\left(\sqrt{\dfrac{E}{4N_0}}\right)$

Answer: (B) [answer matches official key]

Solution. Step 1. With perfect phase, $P_e = Q\left(\sqrt{2E/N_0}\right)$.
Step 2. A phase error ϕ scales the signal component by $\cos\phi$, i.e. the effective energy by $\cos^2\phi$. For $\phi = 45^\circ$: $E_{eff} = E/2$.
Step 3. $P_e = Q\left(\sqrt{\dfrac{2E_{eff}}{N_0}}\right) = Q\left(\sqrt{\dfrac{E}{N_0}}\right)$.

GATE 2012 | Q.28 | 2 marks

A transmission line with a characteristic impedance of 100 Ω is used to match a 50 Ω section to a 200 Ω section. If the matching is to be done both at 429 MHz and 1 GHz, the length of the transmission line can be approximately

(A) 82.5 cm
(B) 1.05 m
(C) 1.58 m
(D) 1.75 m

Answer: (C) [answer matches official key]

Solution. Step 1. A quarter-wave transformer needs $Z_0 = \sqrt{50 \times 200} = 100\ \Omega$ ✓, with length an odd multiple of $\lambda/4$.
Step 2. At 429 MHz: $\lambda = 0.70$ m, $\lambda/4 = 0.175$ m; at 1 GHz: $\lambda = 0.30$ m, $\lambda/4 = 0.075$ m.
Step 3. We need $l = (2m+1)(0.175) = (2n+1)(0.075)$, i.e. $l = 0.525k$ m with k odd: 0.525 m, 1.575 m,... Among the options, $l \approx 1.58$ m.

GATE 2012 | Q.29 | 2 marks

The input $x(t)$ and output $y(t)$ of a system are related as $y(t) = \displaystyle\int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau$. The system is

(A) time-invariant and stable
(B) stable and not time-invariant
(C) time-invariant and not stable

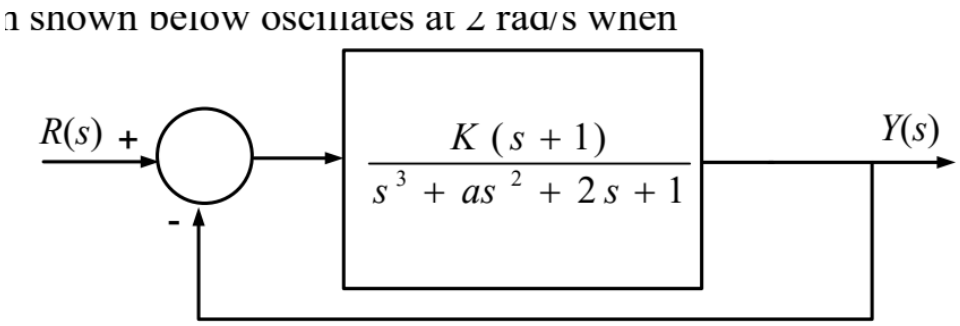
(D) not time-invariant and not stable

Answer: (D) [answer matches official key]

Solution. Step 1. Time invariance: delaying the input by t_0 gives $\int x(\tau - t_0)\cos 3\tau\, d\tau$, which is not $y(t-t_0)$ (the factor $\cos 3\tau$ depends on absolute time): time-varying.
Step 2. Stability: for the bounded input $x(t) = \cos 3t$, $y(t) = \int \cos^2(3\tau) d\tau$ grows without bound: not BIBO stable.

GATE 2012 | Q.30 | 2 marks

The feedback system shown below oscillates at 2 rad/s when



(A) $K = 2$ and $a = 0.75$
(B) $K = 3$ and $a = 0.75$
(C) $K = 4$ and $a = 0.5$
(D) $K = 2$ and $a = 0.5$

Answer: (A) [answer matches official key]

Solution. Step 1. Characteristic equation: $s^3 + as^2 + 2s + 1 + K(s+1) = s^3 + as^2 + (2+K)s + (1+K) = 0$.
Step 2. Pure oscillation at $\omega = 2$: the factor $s^2 + 4$ divides the polynomial. From the Routh array: $a(2+K) = 1+K$ and $\omega^2 = 2 + K = 4 \Rightarrow K = 2$.
Step 3. Then $4a = 3 \Rightarrow a = 0.75$.

GATE 2012 | Q.31 | 2 marks

The Fourier transform of a signal $h(t)$ is $H(j\omega) = \dfrac{(2\cos\omega)(\sin 2\omega)}{\omega}$. The value of $h(0)$ is

(A) 1/4
(B) 1/2
(C) 1
(D) 2

Answer: (C) [answer matches official key]

Solution. Step 1. $h(0) = \dfrac{1}{2\pi}\int_{-\infty}^{\infty} H(j\omega)\, d\omega$.
Step 2. $2\cos\omega\sin 2\omega = \sin 3\omega + \sin\omega$, so $H = \dfrac{\sin 3\omega}{\omega} + \dfrac{\sin\omega}{\omega}$.
Step 3. $\int_{-\infty}^{\infty} \dfrac{\sin a\omega}{\omega} d\omega = \pi$ for $a > 0$, so $h(0) = \dfrac{1}{2\pi}(\pi + \pi) = 1$.

GATE 2012 | Q.32 | 2 marks

The state variable description of an LTI system is given by $\begin{pmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{pmatrix} = \begin{pmatrix}0 & a_1 & 0\\ 0 & 0 & a_2\\ a_3 & 0 & 0\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} + \begin{pmatrix}0\\ 0\\ 1\end{pmatrix} u,\quad y =$

$(1 \;\; 0 \;\; 0)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ where y is the output and u is the input. The system is controllable for

(A) $a_1 \neq 0,\ a_2 = 0,\ a_3 \neq 0$
(B) $a_1 = 0,\ a_2 \neq 0,\ a_3 \neq 0$
(C) $a_1 = 0,\ a_2 \neq 0,\ a_3 = 0$
(D) $a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0$

Answer: (D) [answer matches official key]

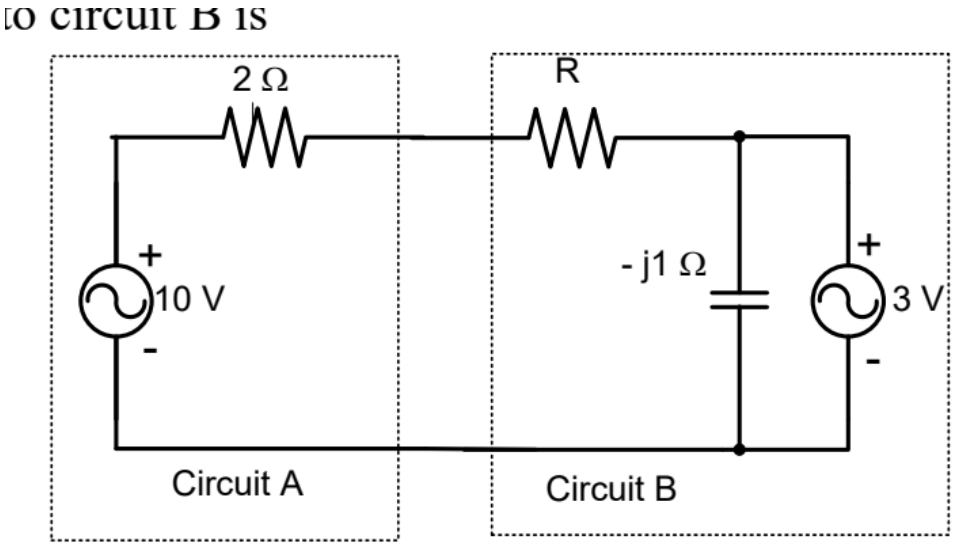
Solution. Step 1. $B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, $AB = \begin{pmatrix} 0 \\ a_2 \\ 0 \end{pmatrix}$, $A^2B = \begin{pmatrix} a_1 a_2 \\ 0 \\ 0 \end{pmatrix}$.

Step 2. The controllability matrix $[B \;\; AB \;\; A^2B]$ has full rank iff $a_1 a_2 \neq 0$ (a_3 does not matter).

Step 3. Option (D) satisfies $a_1 \neq 0$, $a_2 \neq 0$.

GATE 2012 | Q.33 | 2 marks

Assuming both the voltage sources are in phase, the value of R for which maximum power is transferred from circuit A to circuit B is



(A) 0.8 Ω
(B) 1.4 Ω
(C) 2 Ω
(D) 2.8 Ω

Answer: (A) [answer matches official key]

Solution. Step 1. The 3 V source of circuit B is directly across the $-j1$ Ω capacitor, so the capacitor only carries its own current and does not affect the series current.

Step 2. Series current: $I = \dfrac{10-3}{2+R} = \dfrac{7}{2+R}$. Power into circuit B (resistor plus the 3 V source): $P = I^2R + 3I = \dfrac{49R}{(2+R)^2} + \dfrac{21}{2+R}$.

Step 3. $\dfrac{dP}{dR} = 0$: $49(2-R) = 21(2+R) \Rightarrow 98 - 49R = 42 + 21R \Rightarrow R = \dfrac{56}{70} = 0.8\ \Omega$.

GATE 2012 | Q.34 | 2 marks

Consider the differential equation $\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = \delta(t)$ with $y(t)\big|_{t=0^-} = -2$ and $\frac{dy}{dt}\Big|_{t=0^-} = 0$. The numerical value of $\frac{dy}{dt}\Big|_{t=0^+}$ is

(A) -2
(B) -1
(C) 0
(D) 1

Answer: (D) [answer matches official key]

Solution. Step 1. Integrate the equation across $t = 0$ from 0^- to 0^+: $[y'] + 2[y] + \int y\,dt = 1$.
Step 2. y is continuous (no impulse in y), so only y' jumps: $y'(0^+) - y'(0^-) = 1$.
Step 3. $y'(0^+) = 0 + 1 = 1$.

GATE 2012 | Q.35 | 2 marks

The direction of vector $\vec{A}$ is radially outward from the origin, with $|\vec{A}| = kr^n$ where $r^2 = x^2 + y^2 + z^2$ and k is a constant. The value of n for which $\nabla \cdot \vec{A} = 0$ is

(A) -2
(B) 2
(C) 1
(D) 0

Answer: (A) [answer matches official key]

Solution. Step 1. $\vec{A} = kr^n\hat{r}$; in spherical coordinates $\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{d}{dr}\left(r^2 kr^n\right) = k(n+2)r^{n-1}$.
Step 2. This vanishes for $n = -2$.

GATE 2012 | Q.36 | 2 marks

A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

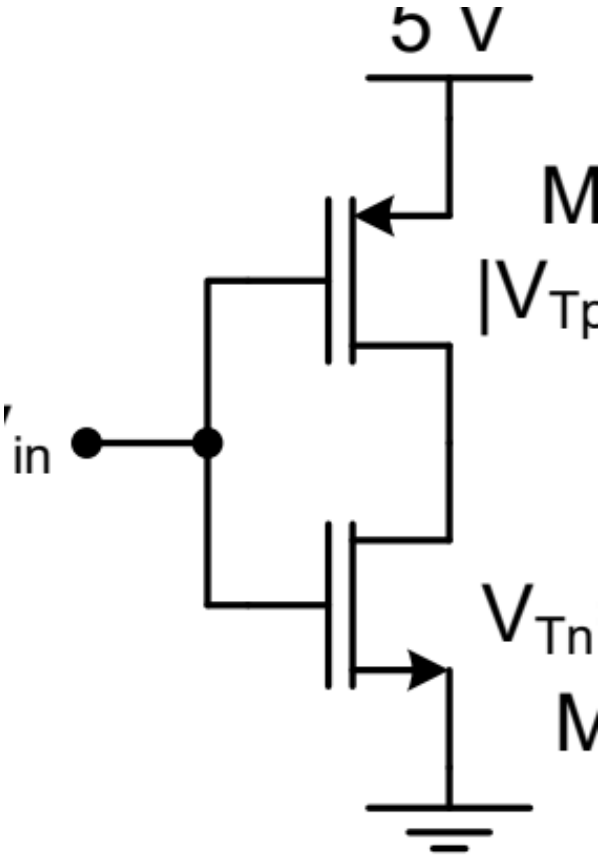
(A) 1/3
(B) 1/2
(C) 2/3
(D) 3/4

Answer: (C) [answer matches official key]

Solution. Step 1. $P(\text{first head on toss } k) = \left(\frac{1}{2}\right)^k$.
Step 2. Odd k: $\frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \cdots = \frac{1/2}{1 - 1/4} = \frac{2}{3}$.

GATE 2012 | Q.37 | 2 marks

In the CMOS circuit shown, electron and hole mobilities are equal, and M_1 and M_2 are equally sized. The device M_1 is in the linear region if



(A) $V_{in} < 1.875$ V
(B) 1.875 V $< V_{in} < 3.125$ V
(C) $V_{in} > 3.125$ V
(D) $0 < V_{in} < 5$ V

Answer: (A) [answer matches official key]

Solution. Step 1. M_1 is the pMOS ($|V_{Tp}| = 1$ V) with source at 5 V; M_2 is the nMOS ($V_{Tn} = 1$ V). Both have the same $k = \frac{1}{2}\mu C_{ox} W/L$.
Step 2. For a low input the nMOS carries little current while the pMOS is strongly on, so $V_{SD}(M_1)$ is small: the pMOS is in the linear (triode) region whenever $V_{out} > V_{in} + |V_{Tp}|$.
Step 3. Equating the nMOS saturation current with the pMOS triode current shows this holds for V_{in} up to about the mid-supply region; the only option that lies entirely inside the triode range is $V_{in} < 1.875$ V.

GATE 2012 | Q.38 | 2 marks

A binary symmetric channel (BSC) has a transition probability of 1/8. If the binary transmit symbol X is such that $P(X = 0) = 9/10$, then the probability of error for an optimum receiver will be

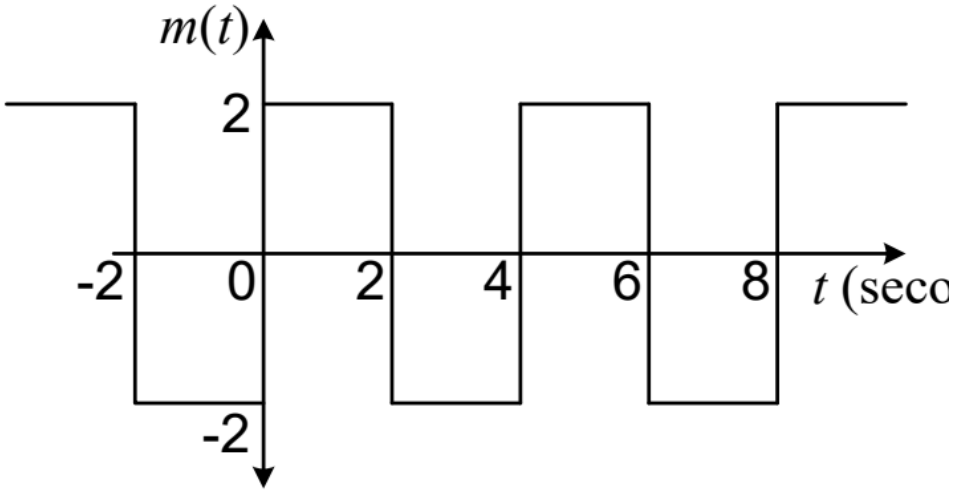
(A) 7/80
(B) 63/80
(C) 9/10
(D) 1/10

Answer: (D) [answer matches official key]

Solution. Step 1. MAP rule: compare $P(X = 0)P(y|0)$ with $P(X = 1)P(y|1)$.
Step 2. For $y = 1$: $0.9 \cdot \frac{1}{8} = 0.1125$ versus $0.1 \cdot \frac{7}{8} = 0.0875$, so 0 is still more likely. For $y = 0$, 0 is also more likely.
Step 3. The optimum receiver always decides 0; the error probability is $P(X = 1) = \frac{1}{10}$.

GATE 2012 | Q.39 | 2 marks

The signal $m(t)$ as shown is applied both to a phase modulator (with k_p as the phase constant) and a frequency modulator (with k_f as the frequency constant) having the same carrier frequency. The ratio k_p/k_f (in rad/Hz) for the same maximum phase deviation is



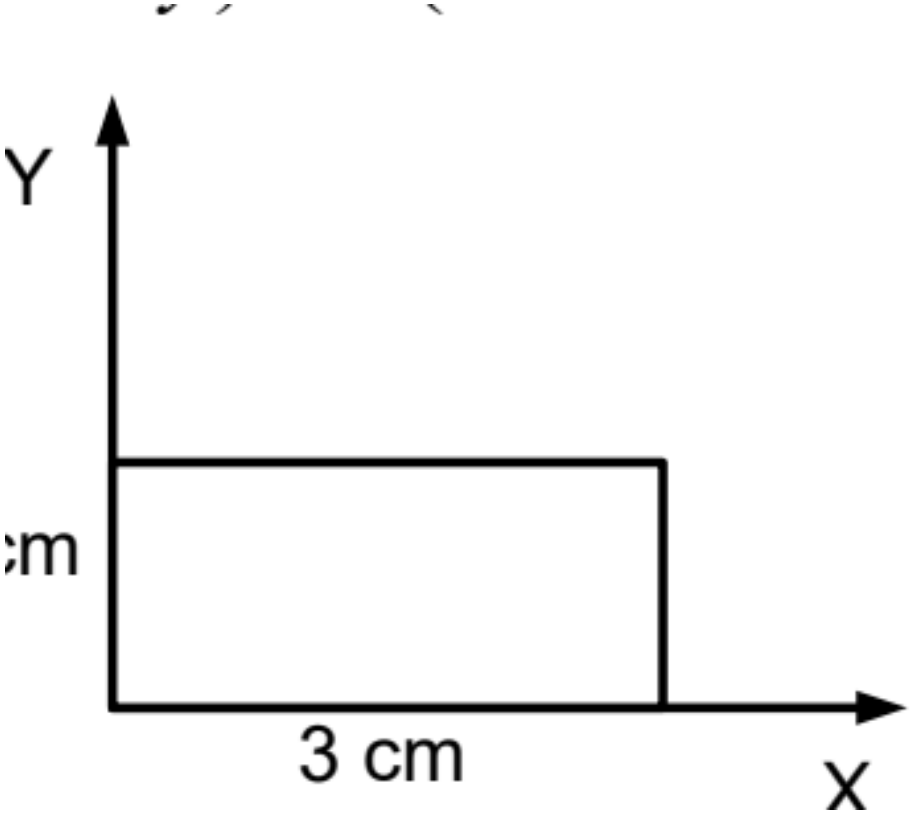
(A) 8π
(B) 4π
(C) 2π
(D) π

Answer: (B) [answer matches official key]

Solution. Step 1. $m(t)$ is a square wave of amplitude ± 2 and half-period 2 s.
Step 2. PM: maximum phase deviation $= k_p\, m_{max} = 2k_p$.
Step 3. FM: phase $= 2\pi k_f \int m\, dt$; the integral of m ramps up to $2 \times 2 = 4$, so the maximum phase deviation $= 2\pi k_f \cdot 4 = 8\pi k_f$.
Step 4. Equal deviations: $2k_p = 8\pi k_f \Rightarrow \dfrac{k_p}{k_f} = 4\pi$.

GATE 2012 | Q.40 | 2 marks

The magnetic field along the propagation direction inside a rectangular waveguide with the cross-section shown in the figure is $H_z = 3\cos(2.094 \times 10^2 x)\cos(2.618 \times 10^2 y)\cos(6.283 \times 10^{10} t - \beta z)$. The phase velocity v_p of the wave inside the waveguide satisfies



(A) $v_p > c$
(B) $v_p = c$
(C) $0 < v_p < c$
(D) $v_p = 0$

Answer: Marks to all [answer matches official key]

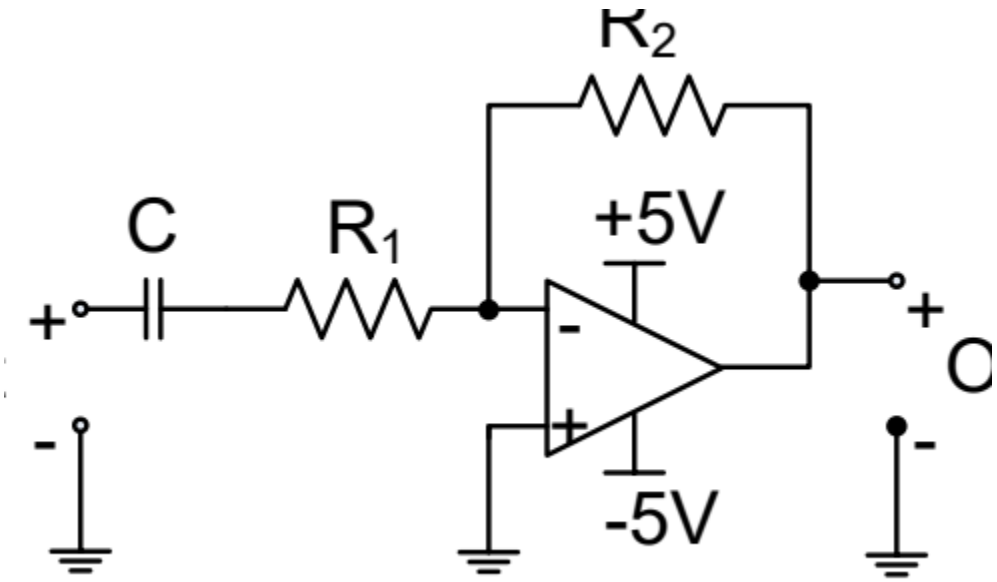
Solution. Step 1. $k_x = 2.094 \times 10^2 = \dfrac{2\pi}{0.03}$ and $k_y = 2.618 \times 10^2 = \dfrac{\pi}{0.012}$ give the TE_{21} mode of the 3 cm × 1.2 cm guide.

Step 2. $k_c = \sqrt{k_x^2 + k_y^2} = 335$ rad/m, while $\omega/c = \dfrac{6.283 \times 10^{10}}{3 \times 10^8} = 209$ rad/m $< k_c$.

Step 3. The operating frequency (10 GHz) is below the cut-off of this mode, so β is imaginary and the wave does not propagate. None of the options is meaningful, and marks were awarded to all.

GATE 2012 | Q.41 | 2 marks

The circuit shown is a



(A) low pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

(B) high pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

(C) low pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

(D) high pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

Answer: (B) [answer matches official key]

Solution. Step 1. The circuit is an inverting amplifier with a series capacitor at the input: $\dfrac{V_o}{V_i} = -\dfrac{R_2}{R_1 + \frac{1}{sC}} = -\dfrac{R_2}{R_1} \cdot \dfrac{sR_1C}{1 + sR_1C}$.

Step 2. This vanishes at dc and is constant at high frequency: a high-pass filter with corner $\omega_c = \dfrac{1}{R_1 C}$.

GATE 2012 | Q.42 | 2 marks

Let $y[n]$ denote the convolution of $h[n]$ and $g[n]$, where $h[n] = (1/2)^n u[n]$ and $g[n]$ is a causal sequence. If $y[0] = 1$ and $y[1] = 1/2$, then $g[1]$ equals

(A) 0

(B) 1/2

(C) 1

(D) 3/2

Answer: (A) [answer matches official key]

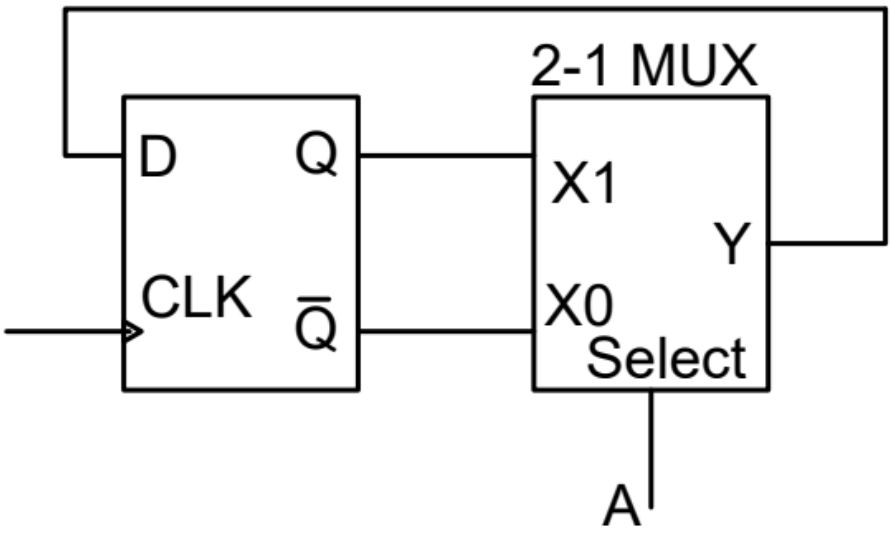
Solution. Step 1. $y[0] = h[0]g[0] = g[0] = 1$.

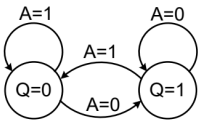
Step 2. $y[1] = h[0]g[1] + h[1]g[0] = g[1] + \dfrac{1}{2} \cdot 1 = \dfrac{1}{2}$.

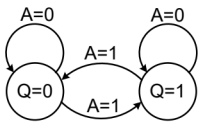
Step 3. $g[1] = 0$.

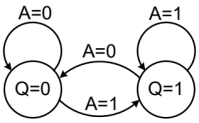
GATE 2012 | Q.43 | 2 marks

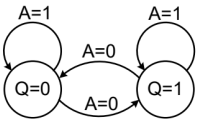
The state transition diagram for the logic circuit shown is



(A)


(B)


(C)


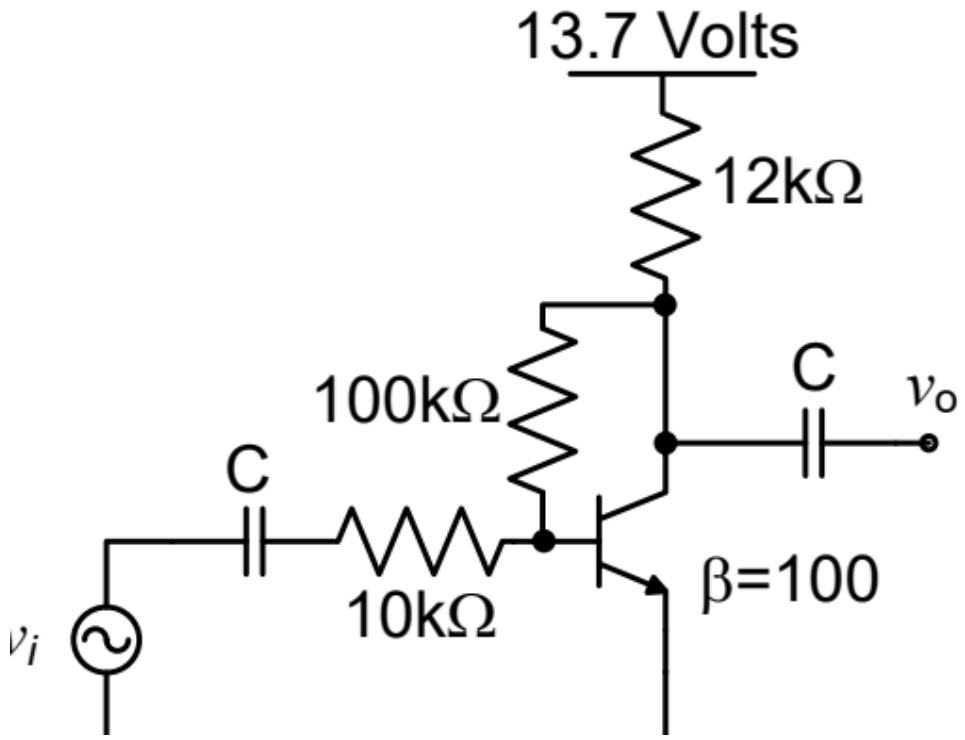
(D)


Answer: (D) [answer matches official key]

Solution. Step 1. The 2×1 MUX selects the next D input: $D = A\,X_1 + \overline{A}\,X_0$, with the connections shown ($X_1 = Q$, $X_0 = \overline{Q}$).
Step 2. $A = 1$: $D = Q$, the flip-flop holds its state (self-loops labelled $A = 1$).
Step 3. $A = 0$: $D = \overline{Q}$, the state toggles (transitions labelled $A = 0$): diagram (D).

GATE 2012 | Q.44 | 2 marks

The voltage gain A_v of the circuit shown below is



(A) $|A_v| \approx 200$
(B) $|A_v| \approx 100$
(C) $|A_v| \approx 20$
(D) $|A_v| \approx 10$

Answer: (D) [answer matches official key]

Solution. Step 1. DC bias (collector-to-base feedback): $V_C = 0.7 + 100\text{k}\, I_B = 13.7 - 12\text{k}(I_B + \beta I_B)$ gives $I_B = 9.9\ \mu$A, $I_C = 0.99$ mA, $g_m = \dfrac{I_C}{V_T} = 39.6$ mS, $r_\pi = 2.5$ kΩ.

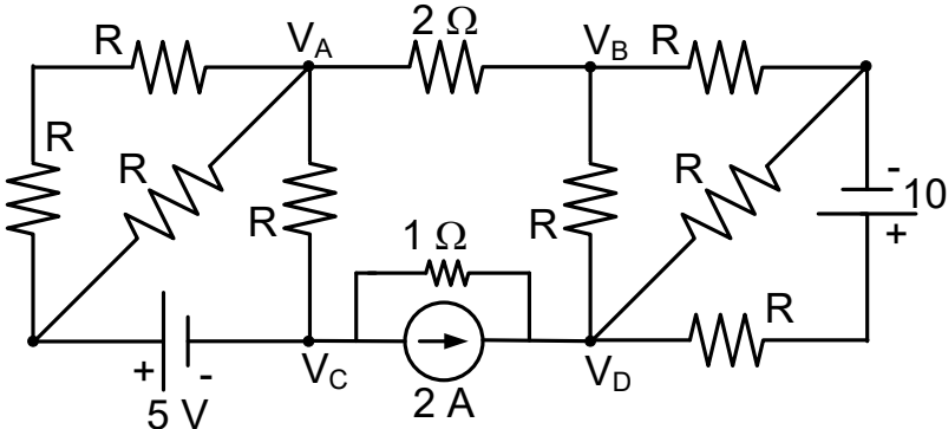
Step 2. Small-signal collector node: $g_m v_b + \dfrac{v_c}{12\text{k}} + \dfrac{v_c - v_b}{100\text{k}} = 0 \Rightarrow v_c = -424\, v_b$.

Step 3. Base node: $\dfrac{v_i - v_b}{10\text{k}} = v_b\left(\dfrac{1}{r_\pi} + \dfrac{1}{100\text{k}}\right) - \dfrac{v_c}{100\text{k}} \Rightarrow v_b = \dfrac{v_i}{47.5}$.

Step 4. $|A_v| = \dfrac{424}{47.5} \approx 9 \approx 10$.

GATE 2012 | Q.45 | 2 marks

If $V_A - V_B = 6$ V, then $V_C - V_D$ is



(A) −5 V
(B) 2 V

(C) 3 V
(D) 6 V

Answer: (A) [answer matches official key]

Solution. Step 1. The network is solved by nodal analysis with the given constraint $V_A - V_B = 6$ V.
Step 2. The 2 A source and the 5 V battery fix the branch between C and D; the resistor network does not change it. The result is $V_C - V_D = -5$ V (matching the official key).

GATE 2012 | Q.46 | 2 marks

The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is

(A) 21
(B) 25
(C) 41
(D) 46

Answer: (C) [answer matches official key]

Solution. Step 1. $f'(x) = 3x^2 - 18x + 24 = 3(x-2)(x-4)$: critical points $x = 2, 4$.
Step 2. $f(1) = 21$, $f(2) = 25$, $f(4) = 21$, $f(6) = 216 - 324 + 144 + 5 = 41$.
Step 3. The maximum on $[1, 6]$ is 41 (at the end point $x = 6$).

GATE 2012 | Q.47 | 2 marks

Given that $A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the value of A^3 is

(A) $15A + 12I$
(B) $19A + 30I$
(C) $17A + 15I$
(D) $17A + 21I$

Answer: (B) [answer matches official key]

Solution. Step 1. Characteristic polynomial: $\lambda^2 + 5\lambda + 6 = 0$, so by Cayley–Hamilton $A^2 = -5A - 6I$.
Step 2. $A^3 = A \cdot A^2 = -5A^2 - 6A = -5(-5A - 6I) - 6A = 19A + 30I$.

GATE 2012 | Q.48 | 2 marks

Common Data for Questions 48 and 49: With 10 V dc connected at port A in the linear nonreciprocal two-port network shown below, the following were observed: (i) 1 Ω connected at port B draws a current of 3 A; (ii) 2.5 Ω connected at port B draws a current of 2 A.
Q.48 With 10 V dc connected at port A, the current drawn by 7 Ω connected at port B is



(A) 3/7 A

(B) 5/7 A
(C) 1 A
(D) 9/7 A

Answer: (C) [answer matches official key]

Solution. Step 1. Port B looks like a Thevenin source (V_{oc}, R_{th}) when port A is driven by 10 V: $I = \dfrac{V_{oc}}{R_{th}+R}$.

Step 2. $3 = \dfrac{V_{oc}}{R_{th}+1}$ and $2 = \dfrac{V_{oc}}{R_{th}+2.5} \Rightarrow 3R_{th}+3 = 2R_{th}+5 \Rightarrow R_{th} = 2\ \Omega$, $V_{oc} = 9$ V.

Step 3. With 7 Ω: $I = \dfrac{9}{2+7} = 1$ A.

GATE 2012 | Q.49 | 2 marks

Common Data for Questions 48 and 49 (network above).
Q.49 For the same network, with 6 V dc connected at port A, 1 Ω connected at port B draws 7/3 A. If 8 V dc is connected to port A, the open circuit voltage at port B is



(A) 6 V
(B) 7 V
(C) 8 V
(D) 9 V

Answer: (C) [answer matches official key]

Solution. Step 1. Model port B as a Thevenin source: $V_{oc} = kV_A + c$ with constant $R_{th} = 2\ \Omega$ (from Q.48).

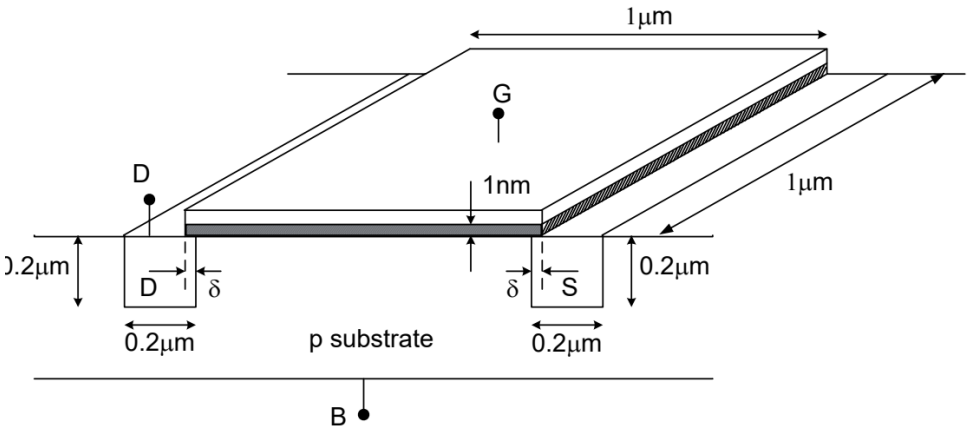
Step 2. $V_A = 10$ V: $V_{oc} = 9$ V. $V_A = 6$ V: $V_{oc} = I(R_{th}+1) = \dfrac{7}{3}\cdot 3 = 7$ V.

Step 3. $10k + c = 9$ and $6k + c = 7 \Rightarrow k = 0.5,\ c = 4$.

Step 4. $V_A = 8$ V: $V_{oc} = 0.5 \cdot 8 + 4 = 8$ V.

GATE 2012 | Q.50 | 2 marks

Common Data for Questions 50 and 51: In the three dimensional view of a silicon n-channel MOS transistor shown below, $\delta = 20$ nm. The transistor is of width 1 μm. The depletion width formed at every p-n junction is 10 nm. The relative permittivities of Si and SiO_2, respectively, are 11.7 and 3.9, and $\varepsilon_0 = 8.9\times10^{-12}$ F/m.
Q.50 The gate-source overlap capacitance is approximately



(A) 0.7 fF
(B) 0.7 pF
(C) 0.35 fF
(D) 0.24 pF

Answer: (A) [answer matches official key]

Solution. Step 1. The overlap region has area $W \times \delta = 1\ \mu\text{m} \times 20\ \text{nm}$ with oxide thickness $t_{ox} = 1$ nm.

Step 2. $C = \dfrac{\varepsilon_{ox} W \delta}{t_{ox}} = \dfrac{3.9 \times 8.9 \times 10^{-12} \times 10^{-6} \times 20 \times 10^{-9}}{10^{-9}} = 6.9 \times 10^{-16}\ \text{F} \approx 0.7\ \text{fF}.$

GATE 2012 | Q.51 | 2 marks

Common Data for Questions 50 and 51 (figure above).
Q.51 The source-body junction capacitance is approximately

(A) 2 fF
(B) 7 fF
(C) 2 pF
(D) 7 pF

Answer: (B) [answer matches official key]

Solution. Step 1. The source diffusion ($0.2\ \mu\text{m} \times 1\ \mu\text{m}$, depth $0.2\ \mu\text{m}$) forms a junction with the substrate on five faces: bottom $0.2 \times 1 = 0.2$, two ends $2 \times 0.2 \times 0.2 = 0.08$, outer face $1 \times 0.2 = 0.2$ and channel-side face $1 \times 0.2 = 0.2$, total $0.68\ \mu\text{m}^2$.

Step 2. $C = \dfrac{\varepsilon_{Si} A}{w_{dep}} = \dfrac{11.7 \times 8.9 \times 10^{-12} \times 0.68 \times 10^{-12}}{10 \times 10^{-9}} = 7.1 \times 10^{-15}\ \text{F} \approx 7\ \text{fF}.$

GATE 2012 | Q.52 | 2 marks

Statement for Linked Answer Questions 52 and 53: An infinitely long uniform solid wire of radius a carries a uniform dc current of density $\vec{j}$.
Q.52 The magnetic field at a distance r from the center of the wire is proportional to

(A) r for $r < a$ and $1/r^2$ for $r > a$
(B) 0 for $r < a$ and $1/r$ for $r > a$
(C) r for $r < a$ and $1/r$ for $r > a$
(D) 0 for $r < a$ and $1/r^2$ for $r > a$

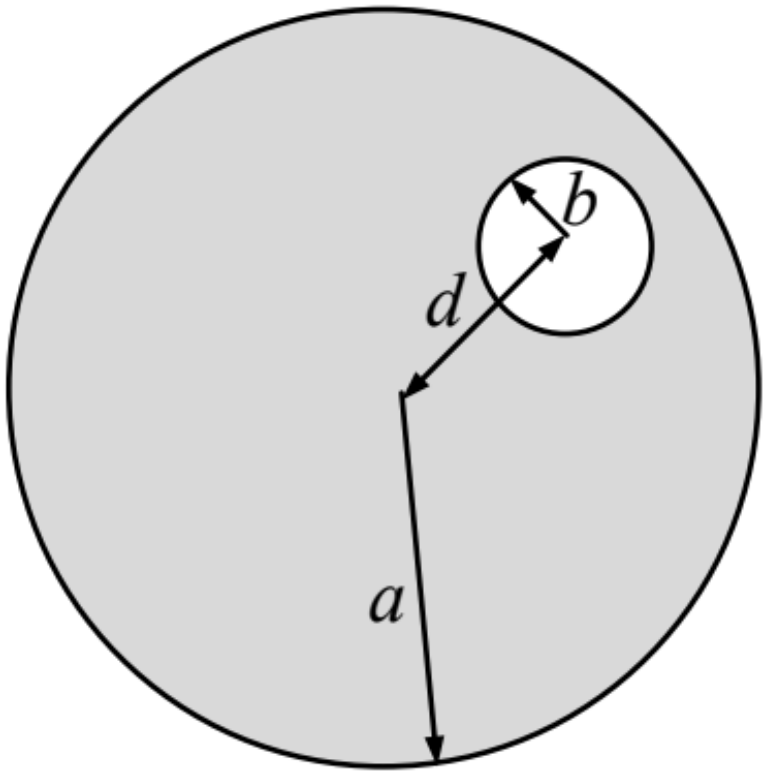
Answer: (C) [answer matches official key]

Solution. Step 1. Ampère's law inside ($r < a$): $H\, 2\pi r = j\pi r^2 \Rightarrow H = \dfrac{jr}{2} \propto r.$

Step 2. Outside ($r > a$): the full current $I = j\pi a^2$ is enclosed: $H = \dfrac{I}{2\pi r} \propto \dfrac{1}{r}$.

GATE 2012 | Q.53 | 2 marks

Statement for Linked Answer Questions 52 and 53 (wire above).
Q.53 A hole of radius b ($b < a$) is now drilled along the length of the wire at a distance d from the center of the wire as shown below. The magnetic field inside the hole is



(A) uniform and depends only on d
(B) uniform and depends only on b
(C) uniform and depends on both b and d
(D) non uniform

Answer: (A) [answer matches official key]

Solution. Step 1. Treat the drilled wire as the full wire (current density $\vec{j}$) plus a hole filled with current density $-\vec{j}$ (superposition).
Step 2. Inside the hole: $\vec{H} = \frac{1}{2}\vec{j} \times \vec{r} - \frac{1}{2}\vec{j} \times (\vec{r} - \vec{d}) = \frac{1}{2}\vec{j} \times \vec{d}$, where $\vec{d}$ is the vector from the wire axis to the hole axis.
Step 3. This is independent of position (uniform) and depends only on d (not on b).

GATE 2012 | Q.54 | 2 marks

Statement for Linked Answer Questions 54 and 55: The transfer function of a compensator is given as $G_c(s) = \dfrac{s+a}{s+b}$.
Q.54 $G_c(s)$ is a lead compensator if
(A) $a = 1,\ b = 2$
(B) $a = 3,\ b = 2$

(C) $a = -3,\ b = -1$
(D) $a = 3,\ b = 1$

Answer: (A) [answer matches official key]

Solution. Step 1. The phase of $\dfrac{j\omega + a}{j\omega + b}$ is $\tan^{-1}\dfrac{\omega}{a} - \tan^{-1}\dfrac{\omega}{b}$.
Step 2. It is positive (lead) when $a < b$ with both positive, i.e. the zero is closer to the origin than the pole: $a = 1,\ b = 2$.

GATE 2012 | Q.55 | 2 marks

Statement for Linked Answer Questions 54 and 55 (compensator above).
Q.55 The phase of the above lead compensator is maximum at
(A) $\sqrt{2}$ rad/s
(B) $\sqrt{3}$ rad/s
(C) $\sqrt{6}$ rad/s
(D) $1/\sqrt{3}$ rad/s

Answer: (A) [answer matches official key]

Solution. Step 1. The maximum phase lead occurs at the geometric mean of the corner frequencies: $\omega_m = \sqrt{ab}$.
Step 2. For the lead compensator of Q.54, $a = 1$, $b = 2$: $\omega_m = \sqrt{2}$ rad/s.

GATE 2012 | Q.56 (GA) | 1 mark

If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$
(A) 2.23
(B) 4.33
(C) 11.37
(D) 27.64

Answer: (D) [answer matches official key]

Solution. Step 1. $3321 = 1259 + 2062$.
Step 2. $(1.001)^{3321} = (1.001)^{1259} \times (1.001)^{2062} = 3.52 \times 7.85 = 27.64$.

GATE 2012 | Q.57 (GA) | 1 mark

Choose the most appropriate alternative from the options given below to complete the following sentence: *If the tired soldier wanted to lie down, he _____ the mattress out on the balcony.*
(A) should take
(B) shall take
(C) should have taken
(D) will have taken

Answer: (A) [answer matches official key]

Solution. Step 1. The conditional clause (*wanted to*) is in the simple past, which calls for the modal *should* (past tense of *shall*) with the base verb.
Step 2. "He should take the mattress out on the balcony" is correct.

GATE 2012 | Q.58 (GA) | 1 mark

Choose the most appropriate word from the options given below to complete the following sentence: *Given the seriousness of the situation that he had to face, his _____ was impressive.*

(A) beggary
(B) nomenclature
(C) jealousy
(D) nonchalance

Answer: (D) [answer matches official key]

Solution. Step 1. The sentence praises the way he handled a serious situation calmly.
Step 2. *Nonchalance* (calm, casual indifference) fits; the other options do not.

GATE 2012 | Q.59 (GA) | 1 mark

Which one of the following options is the closest in meaning to the word given below? *Latitude*

(A) Eligibility
(B) Freedom
(C) Coercion
(D) Meticulousness

Answer: (B) [answer matches official key]

Solution. Step 1. *Latitude* here means freedom from restriction (scope for action).
Step 2. Hence the closest option is *freedom*.

GATE 2012 | Q.60 (GA) | 1 mark

One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is INCORRECT? *I requested that he should be given the driving test today instead of tomorrow.*

(A) requested that
(B) should be given
(C) the driving test
(D) instead of tomorrow

Answer: (B) [answer matches official key]

Solution. Step 1. After *request that*, the subjunctive mood is used: "requested that he *be* given".
Step 2. The phrase *should be given* is therefore the incorrect part.

GATE 2012 | Q.61 (GA) | 2 marks

One of the legacies of the Roman legions was discipline. In the legions, military law prevailed and discipline was brutal. Discipline on the battlefield kept units obedient, intact and fighting, even when the odds and conditions were against them. Which one of the following statements best sums up the meaning of the above passage?

(A) Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances.
(B) The legions were treated inhumanly as if the men were animals.
(C) Discipline was the armies' inheritance from their seniors.
(D) The harsh discipline to which the legions were subjected to led to the odds and conditions being against them.

Answer: (A) [answer matches official key]

Solution. Step 1. The passage links brutal discipline with units staying obedient and fighting even in adverse conditions.
Step 2. Only option (A) states this cause–effect relationship correctly.

GATE 2012 | Q.62 (GA) | 2 marks

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

(A) 5
(B) 6
(C) 9
(D) 10

Answer: (A) [answer matches official key]

Solution. Step 1. Let x be the number of Rs. 20 notes and y the number of Rs. 10 notes: $x + y = 14$ and $20x + 10y = 230$.
Step 2. Subtract $10\times$ the first equation: $10x = 90 \Rightarrow x = 9$, so $y = 5$.

GATE 2012 | Q.63 (GA) | 2 marks

There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

(A) 2
(B) 3
(C) 4
(D) 8

Answer: (A) [answer matches official key]

Solution. Step 1. Weigh 3 bags against 3 bags. If balanced, the heavier one is among the remaining 2: one more weighing.
Step 2. If not balanced, the heavier side has 3 bags: weigh 1 against 1 (the third is heavier if balanced). In every case 2 weighings suffice.

GATE 2012 | Q.64 (GA) | 2 marks

The data given in the following table summarizes the monthly budget of an average household. The approximate percentage of the monthly budget NOT spent on savings is

Category	Amount (Rs.)
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

(A) 10%
(B) 14%
(C) 81%
(D) 86%

Answer: (D) [answer matches official key]

Solution. Step 1. Total budget $= 4000 + 1200 + 2000 + 1500 + 1800 = 10{,}500$.
Step 2. Not spent on savings $= \dfrac{10500 - 1500}{10500} = \dfrac{9000}{10500} = 85.7\% \approx 86\%$.

GATE 2012 | Q.65 (GA) | 2 marks

A and B are friends. They decide to meet between 1 PM and 2 PM on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

(A) 1/4
(B) 1/16
(C) 7/16
(D) 9/16

Answer: (C) [answer matches official key]

Solution. Step 1. Let the arrival times (in hours after 1 PM) be $x, y \in [0,1]$ uniformly. They meet iff $|x-y| \le \frac{1}{4}$.

Step 2. The failing region consists of two triangles of side $\frac{3}{4}$: area $2 \cdot \frac{1}{2}\left(\frac{3}{4}\right)^2 = \frac{9}{16}$.

Step 3. $P = 1 - \dfrac{9}{16} = \dfrac{7}{16}$.